Impact of Free Transport Facility for Women on Their Labour Force Participation in India

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April 2026

Motivation — The Indian FLFP Puzzle

The Core Paradox

- India has grown steadily for three decades, yet urban female labour force participation (FLFP) remains stuck around 18%.
- Rising incomes, rising education — yet women are *leaving*, not entering, the workforce.

Why Transport?

- Transport accounts for $\sim$8.5% of household expenditure in urban India.
- Women's trip chains are complex: paid work + caregiving + shopping = multiple fares, higher cumulative cost, and a gendered 'pink tax'.

Policy Response

- Delhi (2019), Karnataka (2023), Tamil Nadu, Punjab, Telangana — have introduced fare-free or subsidised bus travel for women.
- **Research question:** Do these schemes translate into more women working?

What the Literature Tells Us?

Klasen & Pieters (2015) -The U-Shape & Social Constraints

- U-shaped education–participation curve: low-education women work out of necessity; middle-education women withdraw due to stigma; high-education women re-enter.
- Strong negative income effect: higher husband's income $\rightarrow$ women withdraw. Social norms penalise married, higher-caste, and Muslim women especially.
- Fertility & care burden: more young children $\rightarrow$ lower participation.

Mehrotra & Parida (2017) - Demand Collapse

- Mechanisation displaced $\sim$31 million women from agriculture (2004-12).
- Structural shift to manufacturing/services requires skills many rural women lack, no effective transition from farm to non-farm jobs.
- Limited local job creation: social norms restrict geographic mobility, so lack of nearby work forces withdrawal.

What the Literature Tells Us?

Deshpande & Singh (2021) -Instability, Not Unwillingness

- The problem is not lack of willingness but lack of stable employment.
- Women frequently enter and exit -standard surveys at a point-in-time miss this: 44% participated at least once vs. only 14% measured.
- Discouraged worker effect: high unemployment causes women to stop searching. Male displacement intensifies during job-scarce periods.

Hypothesis - Linking Mobility to Participation

Theoretical Mechanism

- Transport is part of the cost of labour market participation. Removing the fare lowers the monetary barrier to commuting.
- Expanded spatial range $\rightarrow$ better job matching, access to higher-quality work.

Hypothesis

- **H1 (Main):** Fare-free bus schemes increase female labour force participation, particularly at the extensive margin, by reducing the cost of work.
- **H2 (Heterogeneity):** Effects will be largest for low-income women who are transport-constrained, and smallest for moderately-educated women facing structural labour market barriers (Klasen & Pieters 2015).
- **H3 (Limits):** Where labour demand is inadequate, norms are binding, or service quality is poor, mobility gains will be absorbed into domestic work rather than market work (Deshpande & Singh 2021).

Hypothesis - Linking Mobility to Participation

Framing from Literature

- If Mehrotra & Parida (2017) are right that structural demand constraints dominate, free transport alone will not move the needle.
- If Deshpande & Singh (2021) are right about latent supply and discouraged workers, removing cost barriers might restore entry, but only temporarily.

Data - Consumer Pyramids Household Survey (CPHS)

Data Source

- CPHS by Centre for Monitoring Indian Economy (CMIE) — nationally representative longitudinal panel.
- $\sim$170,000 households, interviewed $3\times$ per year $\rightarrow$ high-frequency panel, 33 waves, 2014–2024.
- Sample: women aged 16–60. Final sample: $\sim$3.97 million person-wave observations.

Outcome Variable

- Binary Labour Force Participation (LFP) indicator: 1 if regularly salaried, casually employed, self-employed, unpaid family work, or actively job-seeking.
- 0 if housework, studying, retired, or incapacitated.

Data - Consumer Pyramids Household Survey (CPHS)

Treatment Variable

- Equals 1 if woman resides in a state that implemented a fare-free bus scheme, after the policy introduction date. Otherwise 0.
- Treated states: Delhi, Karnataka, Punjab, Tamil Nadu, Telangana. 71% of observations are from never-treated states (clean control group).

Controls

- Age, education, caste, religion, urban indicator, log household per-capita income, state fixed effects.

Descriptive Patterns

Key Takeaway

- The U-shape means transport policy primarily benefits those at the low end of education — women who work out of necessity, not preference.
- Moderately educated women (the withdrawn middle) need job creation, not just mobility.

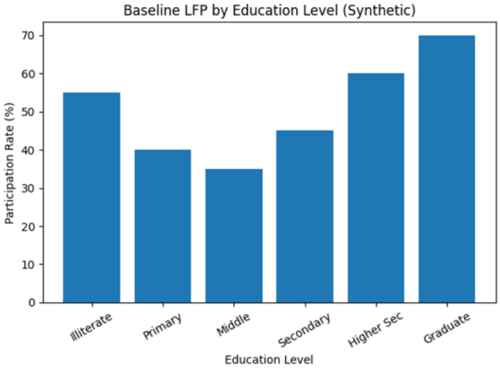

Figure: Baseline Female LFP by Education Level

Empirical Methodology — Staggered DiD

Difference-in-Differences

- Compares change in LFP in policy-adopting states vs. non-adopting states, before and after adoption.
- Key assumption: in the absence of the policy, treated and control states would have followed parallel trends.

Staggered Adoption

- States did not adopt simultaneously; staggered rollout across 2019–2024. This is exploited as the source of identification.
- To avoid bias from already-treated states contaminating the control group, the paper uses the Callaway & Sant'Anna (2021) estimator; only not-yet-treated states serve as controls at each point in time.
- **Baseline DiD:** $\mathrm{LFP}_{ist} = \alpha + \beta\,\mathrm{Policy}_{st} + \gamma\,X'_{ist} + \delta_s + \lambda_t + \varepsilon_{ist}$
- **Event Study:** estimates dynamic effects at each wave relative to adoption ($k = -n$ to $+n$).

Results - Average Treatment Effects

Table 1: Main Estimates — Impact of Free Bus Policy on Female LFP

Callaway-Sant'Anna Staggered DiD Estimator, 2014–2025

	Without Controls		With Controls (IPW)			
	ATT	**SE**	**ATT**	**SE**	**z-stat**	**95% CI**
Overall ATT (Simple Average)	-0.0092***	(0.00098)	-0.0168***	(0.00195)	-8.62	[-0.0206, -0.0130]
Pre-Treatment Average (Pre_avg)	-0.0013***	(0.00025)	-0.0053***	(0.00124)	-4.28	[-0.0078, -0.0029]
Post-Treatment Average (Post_avg)	-0.0112***	(0.00185)	-0.0238***	(0.00410)	-5.79	[-0.0318, -0.0157]
Calendar Time Average (CAverage)	-0.0085***	(0.00131)	-0.0189***	(0.00300)	-6.31	[-0.0248, -0.0131]
OLS (TWFE) — Treated Coefficient	—	—	-0.0515***	(0.00062)	-83.23	[-0.0527, -0.0503]

Results -Event Study

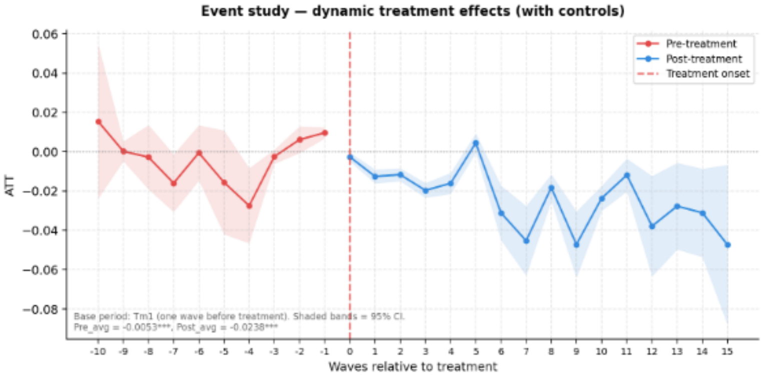

Figure 2: Event study — dynamic treatment effects with 95% confidence intervals

Red = pre-treatment periods; Blue = post-treatment. Vertical dashed line marks treatment onset. Shaded bands = 95% CI.

Figure: Event-Study: Impact of Free Bus Policy on Female LFP

Heterogeneity - Who Is Affected?

Table 3: Heterogeneity Analysis — ATT by Subgroup

Callaway-Sant'Anna Estimator with Individual Controls

Subgroup	ATT	Std. Error	z-statistic	p-value	Sig.
Panel A: Urban/Rural					
Urban	-0.0384***	(0.00900)	-4.26	0.000	***
Rural	-0.0114***	(0.00360)	-3.17	0.002	***
Panel B: Age Group					
Young (16–25 years)	-0.0104***	(0.00384)	-2.71	0.007	***
Prime Working Age (26–35 years)	-0.0068	(0.00431)	-1.58	0.113	—
Middle Age (36–45 years)	-0.0147***	(0.00317)	-4.65	0.000	***
Older Working Age (46–59 years)	-0.0088***	(0.00269)	-3.26	0.001	***
Panel C: Education Level					
Low Education (Up to 10th Std.)	-0.0116***	(0.00173)	-6.69	0.000	***
High Education (Above 10th Std.)	-0.0217***	(0.00469)	-4.62	0.000	***

*Note: *** $p<0.01$, ** $p<0.05$, * $p<0.1$. All specifications control for caste, religion, total income, and relevant demographic variables. Region: 1 = Urban, 0 = Rural. Education: Low = up to 10th standard; High = above 10th standard. Age groups: 1=16-25, 2=26-35, 3=36-45, 4=46-59.*

Evidence from Prior Evaluations — What Does Improve?

Savings & Welfare Gains

- Delhi Pink Ticket (2019): women save **Rs.** 500–600/month (>50% of transport spend). Redirected to food, schooling, healthcare.
- Karnataka Shakti (2023): savings exceed **Rs.** 1,000/month. Major increases in work-related trips reported.
- Tamil Nadu (partial subsidy): **Rs.** 400–500/month savings — modest but meaningful.

Mobility Gains

- Delhi: women's share of bus ridership rose from 33% to 42% by 2022–23.
- Karnataka: 62% of users report additional work-related trips.
- Yet: overcrowding, harassment, unreliable schedules persist — offsetting monetary gains with time and safety costs.

Conclusion & Policy Implications

What We Find

- Fare-free bus schemes generate real welfare gains: lower costs, higher ridership, more autonomous mobility especially for low-income women.
- However, aggregate FLFP does not rise and, by most estimates, modestly declines in treated states relative to counterfactual.
- Structural barriers- occupational segregation, gendered norms, unpaid care burden, lack of suitable jobs - dominate any mobility-driven gains.

Policy Implications

- Complement free transport with service quality investment: expand fleets, improve scheduling, gender-sensitive design (lighting, surveillance, female staff).
- Improve last-mile connectivity: coordinate buses with autos, e-rickshaws, walking infrastructure.
- Targeted subsidies: fiscal costs are high; the largest gains accrue to low-income women - concessional pass systems can be more equitable and sustainable.
- Align with labour market reform: expand female-friendly jobs in clerical, sales, and technical occupations; support flexible work and affordable childcare.
- Monitor rigorously: longitudinal tracking of ridership, savings, employment, and safety incidents — use experimental designs where feasible.